

Daniel Aarao Reis Arturi

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EDUCATION

New York University, Courant Institute | New York City, NY | *M.S. Computer Science* | *Starting Fall 2026*

McGill University | Montreal, Quebec | *B.Sc. Honours Software Engineering* | *Graduated May 2026*

- GPA: 3.99/4.0, Dean's Honour List. Coursework: *Distributed Systems (Grad)*, *Applied ML (Grad)*, *Computer Networks (Grad)*, *Sustainability for ML (Grad)*, *Algorithm Design*, *Operating Systems*, *Database Systems*, *Probability & Statistics*, *Linear Algebra*

RESEARCH & WORK EXPERIENCE

NYU Tandon School of Engineering, VIDA Lab | New York City, NY *Research Scientist* *May 2026 – Present*

- Under PI Claudio Silva on DARPA [expMath \(\\$2.2M\)](#); one of 13 inaugural teams): built the full-stack system placing a Lean 4 theorem-proving agent in mathematicians' hands, a React/Vite UI and in-process FastAPI backend that streams typed proof events over SSE, with git-backed storage and a derived-state SQLite timeline.
- Designed a Chrome extension and local companion server that formalizes labeled LaTeX theorem blocks directly from a live Overleaf document into Lean 4, the program's LaTeX-to-Lean deliverable, with self-repair, per-file progress, and GitHub export.
- Hardened the system for real use: added event-driven live updates, spend controls, and an end-to-end integration test harness, and re-architected proof-state tracking around a single source of truth (git for proof content, SQLite for metadata).

Algoverse | Remote *Research Intern (AI Safety & Interpretability)* *April 2025 – Present*

- Investigating parametric geometry of LoRA-adapted LLMs; demonstrated parameter space convergence across independently fine-tuned misaligned models using Grassmann angle analysis and principal subspace comparison.
- Developed post-hoc vaccination methods that surgically remove misaligned parameter subspaces while preserving reasoning performance (measured on GSM8K); researching multi-thousand turn conversation impacts on model persona stability.

McGill University + University of Passau | Montreal, QC *Research Intern (Graph Database Systems)* *August 2025 – March 2026*

- Designed tree index structures (interval, Dewey, float-range encodings) replacing full graph traversals with fast index lookups; benchmarked query throughput and latency across Neo4j, Kuzu, and Apache AGE.
- Built automated performance testing framework for annotation, regression testing, and cross-system query translation; first-author [arXiv publication](#) under review at EDBT, with [open-source reproducibility package](#). Selected for McGill Science Undergraduate Research Showcase.

McGill University (Distributed Information Systems Group) | Montreal, QC *Research Intern* *January – June 2026*

- Implemented and benchmarked a QUBO approach to distributed data allocation across multiple quantum libraries, running it on D-Wave Advantage hardware and annealing simulators.
- Compared multiple QUBO encodings and annealing-parameter regimes on solution quality and runtime. Quantified the programming, tuning, and execution overhead of quantum versus traditional database evaluation, tracing hardware scaling limits.

PUBLICATIONS & SELECTED PROJECTS

[Shared Parameter Subspaces and Cross-Task Linearity in Emergently Misaligned Behavior](#) [Site] [Code]

- *First Author* | *NeurIPS 2025: UniReps (Oral + Honorable Mention)*, *MechInterp (Spotlight)*, *CogInterp* | *Under review at TMLR*
- Demonstrated cross-task linear structure in emergent misalignment via parametric geometry of LoRA-adapted models; showed that different narrow fine-tuning tasks discover the same shared parameter directions in the weight landscape.

Evaluating a Quantum Solution: How hard can it be? [Code]

- *First Author* | *VLDB 2026: 2nd Workshop on Quantum Computing and Data/Knowledge Management* | *Accepted — presenting*
- Experience report on the practical complexity of building and evaluating database quantum solutions, using QUBO data allocation on D-Wave Advantage hardware and two simulators as a case study; identifies key scaling barriers.

ArxivReader | *Next.js, React, TypeScript, SwiftUI, Supabase* [Live] [Code]

- Full-stack web + native iOS research library for saving, searching, and annotating arXiv papers with reading lists, color-coded tags, author discovery, freeform notes, and real-time cross-platform sync via Supabase with Row Level Security.

Paxos Treasure Island | *Java* [Code]

- Distributed multiplayer game using Paxos for total-order broadcast with ballot-based contention resolution over a custom UDP layer; includes a six-mode failure injection framework with scripted test scenarios and an interactive protocol visualizer.

Energy Efficiency Analysis of ML Architectures | *Python, PyTorch, CodeCarbon*

- Configurable framework for training and energy-profiling ML architectures (BERT, T5, ResNet, ConvNeXt) using CodeCarbon, designed for both local execution and Slurm cluster deployment with HuggingFace Transformers and TorchVision.

NYC Housing Law RAG | *Python, FastAPI, PostgreSQL, pgvector*

- Full-stack RAG system answering NYC housing-law questions with citation-grounded, source-validated answers over official legal corpora; hybrid retrieval, session auth, rate limiting, and deployment runbooks.

AWARDS

McGill Science Undergraduate Research Showcase · NeurIPS UniReps Oral + Honorable Mention · NeurIPS MechInterp Spotlight · Wing Hing Chan Scholarship in Science · Algoverse Merit Scholarship (×2)

TECHNICAL SKILLS

- **Languages:** Python, Java, C/C++, TypeScript, JavaScript, Swift, OCaml, R, SQL, Cypher, Bash/Shell, Assembly, Lean.

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- **ML & AI:** PyTorch, TensorFlow, NumPy, pandas, scikit-learn, LoRA/PEFT, Transformers, CodeCarbon.
- **Web & Tools:** React, Next.js, SwiftUI, Tailwind CSS, Supabase, Vite, Django, Git, LaTeX, Lean 4.
- **Concepts:** Mechanistic Interpretability, AI Safety, Autoformalization, Formal Verification, Multi-Agent Systems, Distributed Computing.
- **Languages:** English (Native), Portuguese (Native), Spanish (Advanced) | *Dual Citizen (Brazil/USA)*.